DATABASE MANAGEMENT SYSTEM — PBL

Library Circulation and Digital Resource Management System

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WHAT IS THE PROJECT?

Introduction

The Library Circulation and Digital Resource Management System is a database-based system designed to manage the complete circulation process of a library. It helps library staff manage printed books, digital resources and member activity in an organised, centralised manner.

The system maintains information about:

- Books & editions
- Authors & publishers
- Categories
- Physical copies
- Members
- Issues & returns
- Reservations
- Fines
- Digital resources
- Digital-access logs

EXISTING PROBLEM

Problem Identification

In many libraries, circulation records are still maintained using manual registers or disconnected spreadsheets. This can lead to:

- Difficulty tracking which copies are available
- Duplicate or inconsistent member records
- Overdue copies going unnoticed
- Errors in manual fine calculation
- Reservation queue order not maintained fairly
- Time-consuming, error-prone report generation

Problem Statement To develop a centralised database management system that efficiently manages books, editions, authors, publishers, categories, copies, members, issues, returns, reservations, fines and digital-access logs.

MAIN OBJECTIVES

Objectives

1. Maintain a centralised database of books, editions and digital resources
2. Manage authors, publishers and categories
3. Track individual copies and their availability
4. Maintain accurate, unique member records
5. Record book issues and returns reliably
6. Enforce a fair reservation queue order
7. Calculate and track fines automatically
8. Log access to digital resources
9. Generate useful circulation reports
10. Improve overall efficiency of library operations

THE SYSTEM WILL COVER

Scope of the Project

Catalogue Management

- Books & editions
- Authors
- Publishers
- Categories

Member Management

- Member registration
- Member details
- Membership type
- Membership status

Circulation Management

- Issues
- Returns
- Reservations
- Fines

Digital Resources & Reporting

- Digital-access logs
- Available titles
- Overdue copies
- Popular books & fine collection

EXISTING APPROACHES VS PROPOSED SYSTEM

Literature Survey

Existing Approach	Limitations
Manual registers	Time-consuming and error-prone
Excel spreadsheets	Data duplication and inconsistency
Separate book / member files	Difficult to integrate or cross-check
Manual overdue tracking	Fines calculated incorrectly
Manual reservation lists	Queue order not maintained fairly

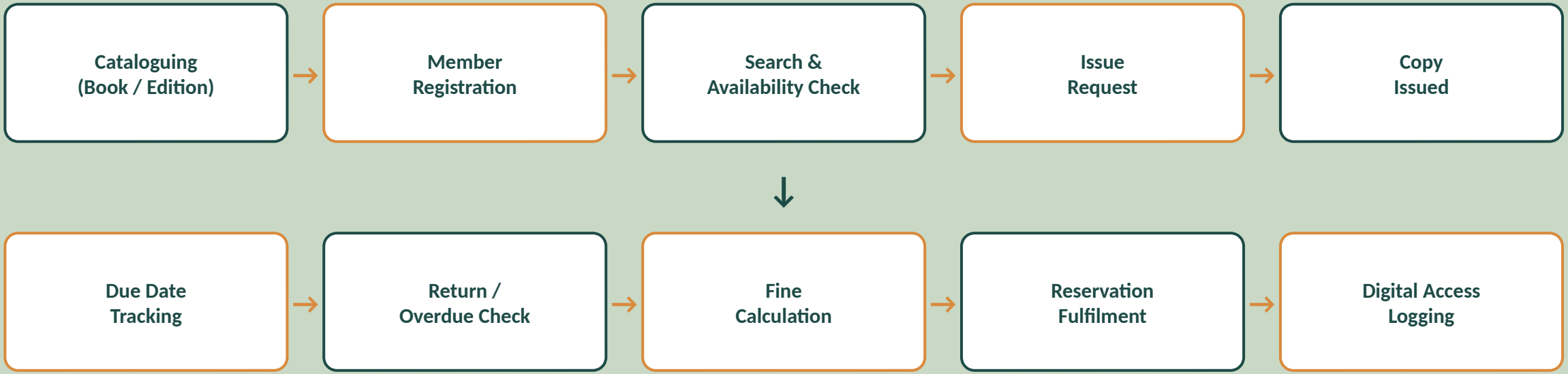
Proposed system provides:

- Centralised data
- Automated fine calculation
- Structured relationships
- Faster searching
- Better data integrity
- Automated reports

Conclusion Existing approaches can manage basic circulation records, but a centralised relational database provides better organisation, consistency and efficient retrieval of library data.

PROPOSED SYSTEM

System Workflow



This slide explains the entire project in one flow — from cataloguing through to return, fine settlement and reservation fulfilment — and is well suited for verbal explanation during the review.

TWO PRIMARY ROLES

Users of the System

1 Librarian / Administrator

- Manage catalogue (books, editions, authors, publishers)
- Manage member records
- Issue and return copies
- Manage reservation queue
- Calculate and record fines
- Generate circulation reports

2 Library Member

- Search the catalogue
- Check copy availability
- Request a reservation
- View issue and due dates
- View fine details
- Access digital resources

CORE ENTITIES IDENTIFIED IN THE PROJECT STATEMENT

Main Entities

- Book
- Edition
- Author
- Publisher
- Category
- Copy
- Member
- Issue
- Return
- Reservation
- Fine
- Digital Resource
- Digital-Access Log

SIMPLIFIED CONCEPTUAL STRUCTURE

ER Diagram

CATEGORY
- Category_ID (PK)
- Category_Name

PUBLISHER
- Publisher_ID (PK)
- Publisher_Name

BOOK
- Book_ID (PK)
- ISBN (unique)
- Category_ID (FK)

AUTHOR
- Author_ID (PK)
- Author_Name

EDITION
- Edition_ID (PK)
- Book_ID (FK)
- Publisher_ID (FK)

MEMBER
- Member_ID (PK)
- Name, Email
- Member_Type

COPY
- Copy_ID (PK)
- Accession_No (unique)
- Edition_ID (FK)

RESERVATION
- Reservation_ID (PK)
- Book_ID (FK)
- Member_ID (FK)

DIGITAL RESOURCE
- Resource_ID (PK)
- Category_ID (FK)

ACCESS LOG
- Log_ID (PK)
- Resource_ID (FK)
- Member_ID (FK)

ISSUE
- Issue_ID (PK)
- Copy_ID (FK)
- Member_ID (FK)

RETURN
- Return_ID (PK)
- Issue_ID (FK)

FINE
- Fine_ID (PK)
- Issue_ID (FK)

Key Relationships

Category → Book / Digital Resource (1:N)
Publisher → Edition (1:N)
Book → Edition (1:N)
Book ↔ Author (M:N)
Edition → Copy (1:N)
Copy → Issue (1:1 active)
Issue → Return / Fine (1:1)
Member → Issue / Reservation / Access Log (1:N)

PROPOSED TABLES, KEYS AND RELATIONSHIPS

Initial Relational Schema

CATEGORY

Category_ID (PK)
Category_Name

PUBLISHER

Publisher_ID (PK)
Publisher_Name
Address

AUTHOR

Author_ID (PK)
Author_Name

BOOK

Book_ID (PK)
Title
ISBN
Category_ID (FK)

EDITION

Edition_ID (PK)
Book_ID (FK)
Publisher_ID (FK)
Year

COPY

Copy_ID (PK)
Edition_ID (FK)
Accession_No
Status

MEMBER

Member_ID (PK)
Name
Email
Member_Type

ISSUE

Issue_ID (PK)
Copy_ID (FK)
Member_ID (FK)
Issue_Date
Due_Date

RETURN

Return_ID (PK)
Issue_ID (FK)
Return_Date

RESERVATION

Reservation_ID (PK)
Book_ID (FK)
Member_ID (FK)
Queue_Pos

FINE

Fine_ID (PK)
Issue_ID (FK)
Amount
Paid_Status

DIGITAL_ACCESS_LOG

Log_ID (PK)
Resource_ID (FK)
Member_ID (FK)
Access_Date

SRS — THE SYSTEM SHOULD ALLOW

Functional Requirements

FR1 Admin can add, update and delete catalogue records (books, editions, authors, publishers).

FR2 Admin can register and manage member records.

FR3 System can maintain multiple physical copies per edition.

FR4 System enforces one active issue per copy at a time.

FR5 Members can search and check availability of titles.

FR6 System can issue and return copies against valid dates.

FR7 System can queue and process reservations in order.

FR8 System can calculate and record non-negative fines.

FR9 System can log access to digital resources.

FR10 System can generate available-titles and overdue reports.

FR11 System can generate member-history and popular-book reports.

FR12 System can generate fine-collection reports.

QUALITY ATTRIBUTES OF THE SYSTEM

Non-Functional Requirements

1

Performance

The system should retrieve catalogue and circulation records efficiently.

2

Security

Only authorised users should be able to modify circulation data.

3

Reliability

The system should maintain accurate, consistent issue and fine records.

4

Usability

The application should have a simple, easy-to-use interface.

5

Scalability

The database should handle a growing catalogue and membership base.

6

Data Integrity

Primary keys, foreign keys and constraints should prevent invalid data.

FIVE PHASES FROM ANALYSIS TO REPORTING

Project Planning

1 → 2 → 3 → 4 → 5

Requirement Analysis

- Identify users
- Identify entities
- Define scope

Database Design

- ER diagram
- Relational schema
- Normalization (3NF)

Database Implementation

- Create tables
- Add constraints
- Insert sample data

Application Development

- Build app (Java/Python)
- Connect to database
- Implement CRUD

Testing & Reporting

- Test database & app
- Generate reports
- Prepare documentation

CONCLUSION

A centralised, structured system for library circulation

The Library Circulation and Digital Resource Management System reduces manual work, improves data consistency, enforces business rules such as unique ISBNs and non-negative fines, and makes circulation information easier to manage and retrieve.

Thank You